

Roll No.

TCS-301

B. TECH. (CSE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022

LOGIC DESIGN

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Define prime implicant and essential prime implicant. Minimize the following Boolean expression by using K-map : (CO1)

$$Y(A, B, C, D) = \sum m(5, 7, 8, 10, 13, 15) + \sum d(0, 1, 2, 3)$$

OR

- (b) A register contains a signed 2's complement number 10001. What will be the content of the register if it is divisible by 5 and then multiplied by 7 ? (CO1)

2. (a) (i) In which base, $121 = 11$ number system is possible ?

- (ii) Determine 8's complement of $(2F0)_{16}$. (CO1)

OR

- (b) Realize $F = (A + B')(CD + E)$ by using (i) NAND gate and (ii) NOR gate. (CO1)

P. T. O.

(2)

3. (a) Explain unit distance code and self-complementing code. Give any *one* example of unit distance code and self-complementing code. (CO1)

OR

- (b) (i) How inversion is achieved by using EX-OR and EX-NOR gate ?
(ii) Implement 2 inputs EX-NOR gate by using minimum number of NAND and OR gate. (CO1)

4. (a) Design circuit of BCD adder. (CO2)

OR

- (b) What is meant by a priority encoder ? How is it different from encoder ?
Draw the circuit of priority encoder. (CO2)

5. (a) Implement $Y(A, B, C, D) = \Pi M(5, 6, 7, 8, 10, 11, 12, 15)$ by using 8 : 1 MUX. (CO2)

OR

- (b) Implement the full sub-tractor circuit using 3 to 8 decoder. (CO2)